

Instructions: Attempt all Questions.

Section A: Multiple Choice Questions (15marks)

Indicate the best answer for these multiple-choice questions.

- 1 Which of the following cannot be achieved by using machine learning?
A: classify respondents into groups based on their response pattern.
B: accurately predict the outcome using supervised learning algorithms.
C: forecast the outcome variable into the future.
D: proving causal relationships between variables
- 2 _____ is the application of machine learning.
A: sentimental analysis
B: traffic prediction
C: speech and face recognition
D: all of the above
- 3 What do we call an application of machine learning methods to large databases?
A: big data computing
B: artificial intelligence
C: data mining
D: internet of Things (IoT)
- 4 Which of the following is the most significant phase in the genetic algorithm?
A: fitness function
B: selection
C: mutation
D: crossover
- 5 Which of the following is TRUE about unsupervised machine learning?
A: unsupervised learning comprises algorithms with no pre-existing outcomes. *un - labelled output*
B: learning algorithms with no control over quality of their predictions.
C: semi-autonomous machine learning where researchers control some parts of the modelling process.
D: fully autonomous machine learning with no human interference
- 6 Dimensionality reduction reduces in _____.
A: collinearity
B: entropy
C: stochastics
D: performance
- 7 Deep learning is a subfield of machine learning where concerned algorithms are inspired by the structured and function of the brain called _____.
A: machine learning
B: artificial neural networks
C: deep learning
D: robotics
- 8 What characterize unlabeled examples in machine learning? There is,
A: no prior knowledge
B: no confusing knowledge
C: prior knowledge
D: plenty of confusing knowledge
- 9 The purpose of regularization in machine learning models is to,
A: increase model complexity
B: decrease model complexity
C: speed up model training
D: increase bias
- 10 The primary goal of feature scaling in machine learning is to,
A: convert categorical features into numerical features
B: normalize the feature values to a standard range
C: create new features from existing ones
D: to reduce the number of features in the dataset
- 11 In a supervised learning problem, the true quantity we really want to minimize is the,
A: training error.

- B: test error
C: generalization error
D: cross-validation error
- 12 The purpose of feature scaling in machine learning is to,
A: remove outliers from the data
B: standardize the range of features
C: increase the complexity of models
D: decrease the dimensionality of features
- 13 In reinforcement learning, a deterministic policy is,
A: a mapping from states to states
B: a mapping from state-action pairs to states
C: a mapping from actions to states
D: a mapping from states to actions
- 14 Which statement about outliers is true?
A: outliers should be identified and removed from a dataset
B: outliers should be part of the training dataset but should not be present in the test set data
C: outliers should be part of the test dataset but should not be present in the training data
D: the nature of the problem determines how outliers are used
- 15 Given a kNN classifier, which one of the following statements is true?
A: The more the number of examples used for classifying an example, the higher the accuracy that will be obtain
B: The more the number of attributes that are used to describe the examples, the more difficult is to obtain high accuracy
C: The most costly part of this method is to learn the model
D: We can use kNN for classification only

Section B: Short and long answer questions (15marks)

1. You have been provided with four different machine learning algorithms that can be used for supervised learning on a given dataset. Explain any four criteria you would use to help decide which one you would use for a given task. (4 marks)
2. Establish the root node using the ID3 algorithm from the data provided in table below. Symptom1 and Symptom2 are symptoms from a Disease, Test is a diagnostic test and the diagnosis is the decision of interest: does the patient have Disease or not. (6 marks)

Case#	Symptom1	Symptom2	Test	Diagnosis
1	Severe	No	negative	No Disease
2	Severe	Yes	negative	Disease
3	Severe	Yes	positive	Disease
4	Mild	No	positive	Disease
5	Mild	Yes	negative	Disease
6	None	No	positive	No Disease
7	Mild	No	negative	No Disease
8	Severe	No	positive	No Disease
9	Mild	Yes	positive	Disease
10	None	Yes	negative	No Disease
11	None	Yes	positive	Disease
12	None	No	negative	No Disease

3. According to road safety reports, road traffic crashes (accidents) are a major cause of death for Kenyans between the ages of five and 70. In addition, many surviving accident victims end up being less productive due to injuries. The leading causes of these crashes include ignorance of traffic rules and human error, largely because of careless riders. Bodabodas contribute a very significant proportion of these crashes. It has been suggested that machine learning can be used to profile bodaboda riders. Suggest one way this can be realized, what data would you require and what processes and machine learning algorithms would you propose? Include your assumptions in your answer. (5marks)

kNN
ID3

classification } supervised
regression
association analysis
clustering

kNN-smaller data set
multi-layer perceptron
ID3
J4E

Decision-node
Leaf node (attribute value)
target